Software Development Organization

2026-2027

1 Project 1

1.1 Context

In a previous project, the Library Management REST-oriented backend service was developed.
This application provides REST endpoints for managing:

- Books
- Genres
- Authors
- Readers
- Lendings

1.2 Problem

The Library Management service has been developed and provides the required functionality, but its software development and delivery process has limitations regarding:

- automation and repeatability of build, testing, integration, and deployment activities;
- systematic assessment of software and test quality;
- early detection of defects and regressions;
- configuration and deployment across different environments;
- continuous integration and continuous delivery/deployment practices.

The development process should therefore evolve towards an automated, repeatable, and measurable CI/CD process that provides timely feedback about the quality and deployability of the software.

1.3 Goals

The goals of this project (P1) are:

- Assess the current development process and system, documenting the relevant *system-as-is*, including:
  - implementation and deployment;
  - existing build and testing practices;
  - test quantity, coverage, and effectiveness;
  - relevant software quality indicators.

- Design the target development and deployment process, documenting the relevant *system-to-be* and the decisions supporting its design.
- Implement an automated CI/CD pipeline that supports:
  - building and packaging the software;
  - static code analysis;
  - automated testing at appropriate levels;
  - test coverage and mutation testing;
  - reporting of build, quality, and test results;
  - creation and management of deployable artifacts;
  - deployment to different environments.
- Assess the existing testing approach, and define an improved automated testing strategy, considering different scopes and levels of testing.
- Assess and improve the automated test suite in terms of test quantity, coverage, representativeness, and effectiveness, adding tests as necessary.
- Support configurable and repeatable deployment of the application across development, staging, and production environments.
- Evaluate and improve the CI/CD process, using collected evidence and appropriate metrics to:
  - compare the initial and improved development processes;
  - identify limitations and bottlenecks;
  - justify improvements introduced throughout the project.

1.4 Technology and Infrastructure Constraints

The following technology and infrastructure constraints apply to the project:

- Build and dependency management: Maven must be used to build the application and manage its dependencies.
- CI/CD: Jenkins must be used to implement the CI/CD pipeline.
- Deployment infrastructure: Virtual servers are provided on premises. Students may alternatively or additionally use other virtual machines or cloud environments available to them, provided that the project requirements are satisfied.

1.5 Conditions

- Number of students per group:
  - 2
  - exceptionally 3 or 1, if accepted by faculty, and implies differences in goals

- First evaluation:
  - Mandatory
  - Submission deadline: 2026-11-01
  - Mode: Individual, synchronous, face-to-face
  - Weight: 100% (or 30% if second evaluation takes places)
- Second evaluation:
  - Optional
  - Submission deadline: (as P2 submission deadline)
  - Mode: Individual, synchronous, face-to-face
  - Weight: 30% (or 0% if it does not take place)

2 Project 2

TBD